

## Differential Equations Lesson 08

### Question 1:

Let  $f(x) = x^2 + 4$  and  $g(x) = x^3 - 3x - 5$

a)  $f(x) + g(x) = \underline{\hspace{2cm}}?$

b)  $D_x[f(x) + g(x)] = \underline{\hspace{2cm}}?$

c)  $D_x[f(x)] + D_x[g(x)] = \underline{\hspace{2cm}}?$

## Question 2:

Let  $f(x) = x^2 - 2$  and  $g(x) = x^3 - x - 1$

a)  $3 \cdot f(x) + 8 \cdot g(x) = \underline{\hspace{2cm}}?$

b)  $D_x[3 \cdot f(x) + 8 \cdot g(x)] = \underline{\hspace{2cm}}?$

c)  $3 \cdot D_x[f(x)] + 8 \cdot D_x[g(x)] = \underline{\hspace{2cm}}?$

d)  $D_x[a \cdot f(x) + b \cdot g(x)] = \underline{\hspace{2cm}}? \cdot D_x[f(x)] + \underline{\hspace{2cm}}? \cdot D_x[g(x)]$

### Question 3:

Let  $f(x) = x^4 + 7$

a)  $3 \cdot f(x) = \underline{\hspace{2cm}}?$

b)  $[3 \cdot f(x)]' = \underline{\hspace{2cm}}? \cdot [f'(x)]$

c)  $[3 \cdot f(x)]'' = \underline{\hspace{2cm}}? [f''(x)]$

d)  $[3 \cdot f(x)]''' = \underline{\hspace{2cm}}? [f'''(x)]$

e)  $[3 \cdot f(x)]^{(n)} = \underline{\hspace{2cm}}? [f^{(n)}(x)]$

f)  $[a \cdot f(x)]^{(n)} = \underline{\hspace{2cm}}? [f^{(n)}(x)]$

### Question 4:

The second-order linear homogeneous differential equation:  $y'' + 5y' + 6y = 0$ .

The homogeneous differential equation has the general form of  $P(x) \cdot y'' + Q(x) \cdot y' + R(x) \cdot y = 0$ .

Here,  $y$  is the unknown function of  $x$ ,  $y'$  and  $y''$  are its first and second derivatives, and  $P(x) = 1$ ,  $Q(x) = 5$ , and  $R(x) = 6$ .

- a) Is  $y_1(x) = C_1 \cdot e^{3x}$  a solution of  $y'' + 5y' + 6y = 0$ ? where  $C_1$  is a constant
- $y_1'(x) = \underline{\hspace{2cm}}?$        $y_1''(x) = \underline{\hspace{2cm}}?$        $y_1''(x) + 5y_1'(x) + 6y_1(x) = \underline{\hspace{2cm}}?$
- b) Is  $y_2(x) = C_2 \cdot e^{2x}$  a solution of  $y'' + 5y' + 6y = 0$ ? where  $C_2$  is a constant
- $y_2'(x) = \underline{\hspace{2cm}}?$        $y_2''(x) = \underline{\hspace{2cm}}?$        $y_2''(x) + 5y_2'(x) + 6y_2(x) = \underline{\hspace{2cm}}?$

c) Is  $y = C_1 \cdot e^{3x} + C_2 \cdot e^{2x}$  a solution of  $y'' + 5y' + 6y = 0$ ?

$$y'(x) = \underline{\hspace{2cm}} \quad y''(x) = \underline{\hspace{2cm}} \quad y''(x) + 5y'(x) + 6y = \underline{\hspace{2cm}}$$

d) Are the functions  $y_3(x) = e^{3x}$  and  $y_4(x) = e^{2x}$  linearly independent?

$$\text{Wronskian determinant} = \begin{vmatrix} y_3(x) & y_4(x) \\ y_3'(x) & y_4'(x) \end{vmatrix} = \underline{\hspace{2cm}}$$

### Question 5:

The second-order linear homogeneous differential equation:  $6y'' - 13y' - 5y = 0$ .

The homogeneous differential equation has the general form of  $P(x) \cdot y'' + Q(x) \cdot y' + R(x) \cdot y = 0$ .

Here,  $y$  is the unknown function of  $x$ ,  $y'$  and  $y''$  are its first and second derivatives,  
and  $P(x) = 6$ ,  $Q(x) = -13$ , and  $R(x) = -5$ .

a) Is  $y_1(x) = C_1 \cdot e^{\frac{5}{2}x}$  a solution of  $6y'' - 13y' - 5y = 0$ ? where  $C_1$  is a constant

$$y_1'(x) = \underline{\hspace{2cm}}?$$

$$y_1''(x) = \underline{\hspace{2cm}}?$$

$$6y_1''(x) - 13y_1'(x) - 5y_1(x) = \underline{\hspace{2cm}}?$$

b) Is  $y_2(x) = C_2 \cdot e^{\frac{-1}{3}x}$  a solution of  $6y'' - 13y' - 5y = 0$ ? where  $C_2$  is a constant

$$y_2'(x) = \underline{\hspace{2cm}}?$$

$$y_2''(x) = \underline{\hspace{2cm}}?$$

$$6y_2''(x) - 13y_2'(x) - 5y_2(x) = \underline{\hspace{2cm}}?$$

c) Is  $y = C_1 \cdot e^{\frac{5}{2}x} + C_2 \cdot e^{\frac{-1}{3}x}$  a solution of  $6y'' - 13y' - 5y = 0$ ?

$$y'(x) = \underline{\hspace{2cm}}?$$

$$y''(x) = \underline{\hspace{2cm}}?$$

$$6y''(x) - 13y'(x) - 5y = \underline{\hspace{2cm}}?$$

d) Are the functions  $y_3(x) = e^{\frac{5}{2}x}$  and  $y_4(x) = e^{2x}$  linearly independent?

$$\text{Wronskian determinant} = \begin{vmatrix} y_3(x) & y_4(x) \\ y_3'(x) & y_4'(x) \end{vmatrix} = \underline{\hspace{2cm}}?$$

## Question 6:

The second-order linear homogeneous differential equation:  $x^2 y'' - 10xy' - 12y = 0$ ,  $x > 0$ .

The homogeneous differential equation has the general form of  $P(x) \cdot y'' + Q(x) \cdot y' + R(x) \cdot y = 0$ .

Here,  $y$  is the unknown function of  $x$ ,  $y'$  and  $y''$  are its first and second derivatives,  
and  $P(x) = x^2$ ,  $Q(x) = -10x$ , and  $R(x) = -12$ .

a) Is  $y_1(x) = C_1 \cdot x^{-1}$  a solution of  $x^2 y'' - 10xy' - 12y = 0$ ? where  $C_1$  is a constant

$$y_1'(x) = \underline{\hspace{2cm}} \quad y_1''(x) = \underline{\hspace{2cm}} \quad x^2 \left[ y_1''(x) \right] - 10x \left[ y_1'(x) \right] - 12 \left[ y_1(x) \right] = \underline{\hspace{2cm}} ?$$

b) Is  $y_2(x) = C_2 \cdot x^{12}$  a solution of  $x^2 y'' - 10xy' - 12y = 0$ ? where  $C_2$  is a constant

$$y_2'(x) = \underline{\hspace{2cm}} \quad y_2''(x) = \underline{\hspace{2cm}} \quad x^2 \left[ y_2''(x) \right] - 10x \left[ y_2'(x) \right] - 12 \left[ y_2(x) \right] = \underline{\hspace{2cm}} ?$$



c) Is  $y = C_1 \cdot x^{-1} + C_2 \cdot x^{12}$  a solution of  $x^2 y'' - 10xy' - 12y = 0$ ?

$$y'(x) = \underline{\hspace{2cm}}?$$

$$y''(x) = \underline{\hspace{2cm}}?$$

$$x^2 y'' - 10xy' - 12y = \underline{\hspace{2cm}}?$$

d) Are the functions  $y_3(x) = x^{-1}$  and  $y_4(x) = x^{12}$  linearly independent?

$$\text{Wronskian determinant} = \begin{vmatrix} y_3(x) & y_4(x) \\ y_3'(x) & y_4'(x) \end{vmatrix} = \underline{\hspace{2cm}}?$$

## Question 7:

The second-order linear homogeneous differential equation:  $y'' + 9y = 0$ .

The homogeneous differential equation has the general form of  $P(x) \cdot y'' + Q(x) \cdot y' + R(x) \cdot y = 0$ .

Here,  $y$  is the unknown function of  $x$ ,  $y'$  and  $y''$  are its first and second derivatives, and  $P(x) = 1$ ,  $Q(x) = 0$ , and  $R(x) = 9$ .

a) Is  $y_1(x) = C_1 \cdot \cos(3x)$  a solution of  $y'' + 9y = 0$ ? where  $C_1$  is a constant

$$y_1'(x) = \underline{\hspace{2cm}}?$$

$$y_1''(x) = \underline{\hspace{2cm}}?$$

$$y_1''(x) - 9y_1(x) = \underline{\hspace{2cm}}?$$

b) Is  $y_2(x) = C_2 \cdot \sin(3x)$  a solution of  $y'' + 9y = 0$ ? where  $C_2$  is a constant

$$y_2'(x) = \underline{\hspace{2cm}}?$$

$$y_2''(x) = \underline{\hspace{2cm}}?$$

$$y_2''(x) - 9y_2(x) = \underline{\hspace{2cm}}?$$

c) Is  $y = C_1 \cdot \cos(3x) + C_2 \cdot \sin(3x)$  a solution of  $y'' + 9y = 0$ ?

$$y'(x) = \underline{\hspace{2cm}}?$$

$$y''(x) = \underline{\hspace{2cm}}?$$

$$y'' + 9y = \underline{\hspace{2cm}}?$$

d) Are the functions  $y_3(x) = \cos(3x)$  and  $y_4(x) = \sin(3x)$  linearly independent?

$$\text{Wronskian determinant} = \begin{vmatrix} y_3(x) & y_4(x) \\ y_3'(x) & y_4'(x) \end{vmatrix} = \underline{\hspace{2cm}}?$$

## Question 8:

The second-order linear homogeneous differential equation:  $x^2 y'' - 9x \cdot y' + 25y = 0$ ,  $x > 0$ .

The homogeneous differential equation has the general form of  $P(x) \cdot y'' + Q(x) \cdot y' + R(x) \cdot y = 0$ .

Here,  $y$  is the unknown function of  $x$ ,  $y'$  and  $y''$  are its first and second derivatives,

and  $P(x) = x^2$ ,  $Q(x) = -9x$ , and  $R(x) = 25$ .

a) Is  $y_1(x) = C_1 \cdot x^5$  a solution of  $x^2 y'' - 9x \cdot y' + 25y = 0$ ? where  $C_1$  is a constant

$$y_1'(x) = \underline{\hspace{2cm}} \quad y_1''(x) = \underline{\hspace{2cm}} \quad x^2 \left[ y_1''(x) \right] - 9x \left[ y_1'(x) \right] + 25y_1(x) = \underline{\hspace{2cm}}$$

b) Is  $y_2(x) = C_2 \cdot x^5 \cdot \ln(x)$  a solution of  $x^2 y'' - 9x \cdot y' + 25y = 0$ ? where  $C_2$  is a constant

$$y_2'(x) = \underline{\hspace{2cm}} \quad y_2''(x) = \underline{\hspace{2cm}} \quad x^2 \left[ y_2''(x) \right] - 9x \left[ y_2'(x) \right] + 25y_2(x) = \underline{\hspace{2cm}}$$

c) Is  $y = C_1 \cdot x^5 + C_2 \cdot x^5 \cdot \ln(x)$  a solution of  $x^2 y'' - 9x \cdot y' + 25y = 0$ ?

$$y'(x) = \underline{\hspace{2cm}}?$$

$$y''(x) = \underline{\hspace{2cm}}?$$

$$x^2 y'' - 9x \cdot y' + 25y = \underline{\hspace{2cm}}?$$

d) Are the functions  $y_3(x) = x^5$  and  $y_4(x) = x^5 \cdot \ln(x)$  linearly independent?

$$\text{Wronskian determinant} = \begin{vmatrix} y_3(x) & y_4(x) \\ y_3'(x) & y_4'(x) \end{vmatrix} = \underline{\hspace{2cm}}?$$

## Question 9:

The second-order linear homogeneous differential equation:  $x^2 y'' - 6x \cdot y' + 12y = 4x + 5$ ,  $x > 0$ .

a) The equation  $x^2 y'' - 6x \cdot y' + 12y = 0$  is second-order linear homogeneous differential equation.

Is  $y_c(x) = C_1 \cdot x^3 + C_2 \cdot x^4 \cdot \ln(x)$  a solution of  $x^2 y'' - 6x \cdot y' + 12y = 0$ ?

$y_c(x)$  is called the complementary solution.

$$y_c'(x) = \underline{\hspace{2cm}}?$$

$$y_c''(x) = \underline{\hspace{2cm}}?$$

$$x^2 [y_c''(x)] - 6x \cdot [y_c'(x)] + 12[y_c(x)] = \underline{\hspace{2cm}}?$$

b) Is  $y_p(x) = \frac{2}{3}x + \frac{5}{12}$  a solution of  $x^2 y'' - 6x \cdot y' + 12y = 4x + 5$ ?

$y_p(x)$  is called the particular solution.

$$y_p'(x) = \underline{\hspace{2cm}}?$$

$$y_p''(x) = \underline{\hspace{2cm}}?$$

$$x^2 [y_p''(x)] - 6x \cdot [y_p'(x)] + 12[y_p(x)] = \underline{\hspace{2cm}}?$$

c) Note:  $y_c(x) + y_p(x) = C_1 \cdot x^3 + C_2 \cdot x^4 + \frac{2}{3}x + \frac{5}{12}$ .

Is  $y_c(x) + y_p(x)$  a solution of  $x^2 y'' - 6x \cdot y' + 12y = 4x + 5$ ?

$$\left[ y_c(x) + y_p(x) \right]' = y_c'(x) + y_p'(x) = \underline{\hspace{2cm}}?$$

$$\left[ y_c(x) + y_p(x) \right]'' = y_c''(x) + y_p''(x) = \underline{\hspace{2cm}}?$$

$$x^2 \left[ y_c(x) + y_p(x) \right]'' - 6x \cdot \left[ y_c(x) + y_p(x) \right]' + 12 \left[ y_c(x) + y_p(x) \right] = \underline{\hspace{2cm}}?$$

